



Pundarikaksh Narayan Tripathi

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Education

Babu Banarasi Das University

B.Tech in Computer Science and Engineering (Artificial Intelligence)

Lucknow, UP

Expected May 2027

- **Relevant Coursework:** Machine Learning, Data Science, Artificial Neural Networks, Computer Networking, Database Management Systems, Design and Analysis of Algorithms.

IIT Mandi

Minor in Artificial Intelligence

Mandi, HP

March 2025 - Present

- **Relevant Coursework:** Programming in Python, Linear Algebra, Calculus, Statistics, Probability, Traditional AI and Machine Learning, Deep Learning for Computer Vision

Projects

Aether (In Development) | C++20, Go, Distributed Systems

- Architecting a distributed multimodal search engine utilizing a Go/Redis asynchronous crawling cluster.
- Designing the C++ core for BM25 inverted-indexing and HNSW vector retrieval, integrating a custom Vision-Language Model (VLM) inference backend via gRPC.

VoltaSplat (Active Research) | C++20, CUDA C++, PyTorch

- Developing a differentiable CUDA rasterization engine from scratch for 3D Gaussian Splatting, extending PyTorch via custom C++ bindings.
- Engineering parallelized forward/backward pass kernels, utilizing NVIDIA CUB for tile-based radix sorting and hardware-accelerated alpha compositing.

TernixEngine (In Development) | C++20, AVX2 SIMD, HPC

- Building a dependency-free C++ inference backend specifically optimized for 1.58-bit ternary Vision-Language Models.
- Replacing standard FP16 matrix multiplications with AVX2 SIMD-accelerated integer additions to drastically reduce memory footprint and maximize token throughput.

Caissa - Chess Engine (In Development) | C, Algorithmic Optimization, Machine Learning

- Built a dependency-free chess engine from scratch featuring custom Alpha-Beta search algorithms, transposition tables, and Magic Bitboards.
- Integrated machine learning via Texel Tuning (gradient descent) to optimize static evaluation heuristics and maximize node evaluation throughput.

LumaSort-Engine | C++20, OpenGL Compute Shaders, OpenCV

- Engineered a high-performance, real-time computer vision engine utilizing OpenGL Compute Shaders for massively parallel pixel processing.
- Implemented global luminance-based 1D optimal transport and fluid dynamics (flow fields) to calculate deterministic pixel trajectories at high frame rates.

Technical Skills

Languages: C, C++ (C++17/20), CUDA C++, Python, Go, SQL

Technologies:

– **AI & Machine Learning:** PyTorch, scikit-learn, ONNX, Hugging Face, OpenCV, Quantization (1.58-bit / GGUF)

– **Data, MLOps & Infrastructure:** Kafka, Apache Spark, Airflow, Docker, Kubernetes (K8s), MLflow, AWS SageMaker, CMake

– **Backend & Systems:** gRPC, Redis, PostgreSQL, Node.js, Hardware Intrinsics (SIMD/AVX2), Microservices, Memory Management

Concepts: Artificial Intelligence, High-Performance Computing (HPC), GPU Architecture, System Design, Data Engineering

Achievements

- Secured **2nd Place at HackArena 2025** for **CivicAgent**. Engineered the core AI routing and deterministic backend architecture to autonomously triage civic issues without manual intervention.

Social Engagements & Leadership

Campus Partner: Perplexity AI (Fall '25 Cohort, Aug 2025 – Jan 2026)

Member: Google Developer Groups (GDG) Lucknow (Aug 2025 – Present)